

## Exercise 1

## SQL Fundamentals

### Question 1

Select \*  
FROM employees;

| id | first-name | last-name | department | Salary | hire-date  | city         |
|----|------------|-----------|------------|--------|------------|--------------|
| 1  | Alice      | Green     | IT         | 70000  | 2020.01.10 | Johannesburg |
| 2  | Brian      | Lee       | HR         | 45000  | 2019.03.22 | Cape Town    |
| 3  | Cathy      | Zulu      | Finance    | 65000  | 2018.07.18 | Durban       |
| 4  | David      | Mobeni    | Marketing  | 50000  | 2021.11.05 | Pretoria     |
| 5  | Eva        | Naidoo    | IT         | 72000  | 2017.09.30 | Johannesburg |

### Question 2

Select DISTINCT department  
FROM employees;

| department |
|------------|
| IT         |
| HR         |
| finance    |
| Marketing  |

### Question 3

Select first-name, last-name, Salary  
FROM employees  
ORDER BY Salary DESC;

| first-name | last-name | salary |
|------------|-----------|--------|
| Eva        | naldo     | 72000  |
| Alice      | Green     | 70000  |
| Cathy      | Zulu      | 65000  |
| David      | mokoena   | 50000  |
| Brian      | Lee       | 45000  |

#### Question 4

```

SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

```

| id | first-name | last-name | Salary |
|----|------------|-----------|--------|
| 5  | Eva        | naldo     | 72000  |
| 1  | Alice      | Green     | 70000  |
| 3  | Cathy      | Zulu      | 65000  |

#### Question 5

```

SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';

```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 5  | Eva        | naldo     | IT         |

### Question 6

```
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

| id | first-name | last-name | department | salary |
|----|------------|-----------|------------|--------|
| 3  | Cathy      | Zulu      | Finance    | 65000  |

### Question 7

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR
      department = 'marketing';
```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 4  | David      | Mokoena   | marketing  |

### Question 8

```
SELECT id, first-name, last-name, department
FROM employees
WHERE NOT department = 'IT';
```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 4  | David      | Mokoena   | marketing  |

### Question 9

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'finance');
```

| id | first-name | last-name | department |
|----|------------|-----------|------------|
| 1  | Alice      | green     | IT         |
| 2  | Brian      | Lee       | HR         |
| 3  | Cathy      | Zulu      | finance    |
| 5  | Eva        | naidoo    | IT         |

### Question 10

```
SELECT id, first-name, last-name, department, salary, city
FROM employees
WHERE department = 'IT'
      AND salary > 65000
      AND city = 'Johannesburg';
```

| id | first-name | last-name | department | Salary | city         |
|----|------------|-----------|------------|--------|--------------|
| 1  | Alice      | green     | IT         | 70000  | Johannesburg |
| 5  | Eva        | naidoo    | IT         | 72000  | Johannesburg |